Veer Phatak

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PROFILE

Final-year MEng Aeronautics and Astronautics student specialising in Aerodynamics at the University of Southampton, graduating July 2027 with a 71.7% first-class Year 3 average. Team leader and aerodynamic design lead on a final-year project investigating the cornering aerodynamics of a MotoGP-style motorcycle through CFD and wind tunnel testing. Three seasons of Formula Student race engineering, a Formula 1 lap time and race strategy model validated against telemetry, a first-class experimental fluid dynamics project with a bespoke computer-vision measurement system, and an R&D internship at TREND Networks where a redesigned part was approved for manufacture.

CORE SKILLS

Aerodynamics and CFD: ANSYS Fluent, XFLR5, XFOIL, OpenVSP/VSPAERO, panel methods (Python), aerofoil and boundary layer analysis, aerodynamic trade studies

Wind tunnel and experimental: wind tunnel testing, CFD and experiment correlation, test rig design and build, calibration, computer-vision measurement, uncertainty analysis

Modelling and data: Python (NumPy, pandas, SciPy, OpenCV, Matplotlib), MATLAB, JavaScript, Excel, Git; lap time simulation, least-squares fitting, Monte Carlo, Kalman filtering

CAD and manufacture: SolidWorks, design for manufacture (injection moulding, additive manufacturing), rapid 3D-printed prototyping (Bambu Studio), fit and tolerance checks

FEA and motorsport: ANSYS Mechanical, mesh convergence studies; MoTeC i2, setup evaluation, race strategy modelling

ENGINEERING PROJECTS

Final-Year Group Design Project: Cornering Aerodynamics of a MotoGP-Style Motorcycle Sep 2026 - Present

Supervisors: Dr David Marshall and Dr Renan Soares

- **Team leader** of a student group investigating how drag, downforce, side force and the yaw and roll moments vary with lean angle, through CFD and wind tunnel testing; coordinating the team and liaising with supervisors.
- **Aerodynamic design lead**, responsible for aerodynamic direction across the CFD and wind tunnel programme.
- **CAD development and manufacturing lead**, responsible for the CAD model and manufacture of the wind tunnel test model.

Formula 1 Lap Time and Race Strategy Model (personal project, runs in the browser) 2026

- Built a quasi-steady lap time model in Python on circuit geometry derived from car position telemetry, predicting the 2025 Bahrain pole lap to **within about 1%** (90.36 s against 89.84 s) and holding that accuracy on sections of track it was not fitted to.
- Fitted tyre degradation, fuel and driver effects by least squares across **979 racing laps**, measured pit loss from 28 green-flag stops, and ranked **14,532 legal strategies**, placing the winning strategy 3.8 s from the calculated optimum.
- Extended it to a Monte Carlo treatment of strategy under safety cars, pit stop variation and tyre scatter, and ported the solver to JavaScript so it can be re-run interactively at veerphatak.github.io.

Individual Master's Project: Hydrodynamic Analysis of Large Plastics in Open-Channel Flow Oct 2025 - May 2026

Graded 75% (First). Supervisor: Dr Sergio Maldonado

- Designed and built a laboratory open-channel flume and paddlewheel drive (SolidWorks, Arduino stepper control) to reproduce turbulent flow, covering component selection, bearing and shaft mounting, sealing and camera illumination.
- Developed a Python/OpenCV measurement system (segmentation, contour fitting, Kalman filtering, homography calibration) reconstructing trajectories from 240 fps footage at **98.4% mean detection coverage and 1.5 mm positional accuracy**.
- Analysed **58 experimental runs** across spheres, rods and discs using velocity, slip, drag, dispersion and tortuosity, with Reynolds and Stokes number analysis comparing flume and water tunnel results.
- Validated every tracked run against raw footage to reject reflections and missed detections; roughly 9,400 lines of published Python.

Fixed-Wing UAV Wing Design, Aerodynamics and Stability Lead (Group Project) 2024 - 2025

- **Ranked 1st in the cohort:** delivered the largest minimum-to-maximum speed range in the year group, **23.1 m/s predicted in XFLR5 against 19.8 m/s measured in flight**, using single-slot Fowler flaps and vortex generators.
- Designed the box-spar wing structure over three test-driven iterations on a purpose-built bending and torsion rig, and compared planforms in XFLR5 and XFOIL within mass and volume limits.
- Led configuration selection using decision matrices, and managed Gantt and resource planning that delivered the project early.

ENGINEERING EXPERIENCE

TREND Networks, R&D Intern Jul 2026 - Sep 2026

- **Battery compartment redesign:** redesigned an existing product's battery compartment for user-replaceable batteries under EU regulations within a fixed envelope, balancing battery protection, durability, styling and ease of use. Ten iterations produced a design **approved and released for manufacture**.
- **Handheld instrument enclosure:** designed from initial concept around a pre-manufactured PCB, fitting clearances, port positions and mounting points while reconciling requirements from several engineers, over three iterations.
- **Ribbon fibre spool:** designed a spool to feed and protect fibre during testing, maximising stored fibre length within each tester case over three iterations.
- Ran all office 3D printing, prototyping each iteration the same day to check fit, wall thickness, warping and tolerances against the CAD, with injection moulding in mind for production parts.

Southampton University Formula Student (SUFST), Race Engineering and Chassis 2023 - Present

- Built Python and MATLAB lap-time and tyre degradation simulations and analysed logged data in MoTeC i2 to evaluate setup trade-offs, contributing to **1st in the FSUK Simulator Championship 2026**, and to 2nd in the Simulator Championship and 10th overall at FSUK 2025.
- Researched front roll hoop integration into the chassis and presented technical and feasibility findings to the department and team management.

EDUCATION

University of Southampton, MEng Aeronautics and Astronautics (Aerodynamics) Sep 2023 - Jul 2027

- **Year 3: 71.7% average (First).** Aerospace Control Design 80%, Individual Project 75%, Aircraft Structural Design 75%, Advanced Aeronautics 74%, Aerothermodynamics 68%.
- **Wind tunnel laboratories:** measured NACA 0020 pressure distribution against angle of attack, boundary layer development and finite wing lift and drag, and correlated results against CFD, XFOIL and XFLR5.
- **Advanced Aeronautics:** implemented a Python doublet panel method with accuracy and panel-distribution convergence studies.
- **FEA:** ANSYS Mechanical mesh convergence study from roughly 24,000 to 1.9 million elements, balancing peak stress accuracy against solve time and memory.

Reading School, Reading Sep 2016 - Jul 2023

- A-levels: A* Economics, A Mathematics, A Physics, B Classical Civilisation. AS Additional Mathematics (A). 10 GCSEs including eight grade 9s.

LEADERSHIP AND WORK EXPERIENCE

Head Coach, Boyne Hill Cricket Club May 2025 - Present

- Plan and lead weekly two-hour training sessions for U9 girls and U13 boys teams.

Other experience: Private Tutor, ages 8 to 14 (2022 - 2025); Front of House, Seasonality Restaurant (2022 - 2024); Dynamos Cricket Coach, ECB programme (2021 - 2022); Tutor, Brilliant Tutors Academy (2022); Sales Assistant and Till Manager, Kay's Chemist (2021 - 2022).

ACHIEVEMENTS AND INTERESTS

- 1st overall, FSUK Simulator Championship 2026; 2nd in the FSUK lap time challenge and Simulator Championship 2025.
- Boyne Hill CC 1st XI wicketkeeper-batter with 16 years of competitive cricket; captained Southampton University CC indoor team and outdoor 3rd XI.
- ECB and Berkshire qualified cricket coach; qualified first aider; Sports Prefect at Reading School.

References available on request.